# CARMEL PUBLIC SCHOOL

# VAZHAKULAM




## SENIOR SECONDARY CERTIFICATE EXAMINATION

## 2026-2027

## RECORD OF PROJECT WORK

## IN

## COMPUTER SCIENCE

## DONE BY

**NAME** :

**REGISTER NUMBER** :

# CARMEL PUBLIC SCHOOL

# VAZHAKULAM




## CERTIFICATE

This is to certify that is the Bonafide Record of the Project Work of _ _ _

# ACKNOWLEDGEMENT

It is with immense gratitude that I acknowledge the guidance, encouragement, and support received from many individuals in the successful completion of my project work.

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# CAR SHOWROOM MANAGEMENT SYSTEM

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# INTRODUCTION

The Car Showroom Database Management System is a menu-driven application developed in Python with a MySQL database. It supports common showroom operations by storing and managing customer details, vehicle inventory, staff information, and transaction records. When started, the program connects to a local MySQL server and creates the database tables if they do not already exist.

Users can add, view, update, and delete customer and inventory records, while staff-management and administrative functions are restricted to Managers. The transaction workflow supports creating quotes and finalized sales, checking available stock, reserving, or selling a vehicle, cancelling a quote, and printing or reprinting an invoice. Transaction listings show related car, customer, and employee names for easier reference.

The application also provides showroom statistics, quarterly sales and profit summaries, staff sales performance, and customer transaction history. It is intended as an educational project demonstrating Python functions, input validation, SQL operations, and Python–MySQL connectivity; its sample credentials and plain-text passkeys make it unsuitable for real customer or production use.

# HARDWARE AND SOFTWARE SPECIFICATIONS

## Hardware

- Processor : Dual-core
- Processor Speed : 1 GHz or faster
- RAM : 4 GB RAM or Above
- Storage : 1 GB of free disk space for Application & local MYSQL DB
- Peripherals : Keyboard, monitor, and mouse or touchpad

## Software

- Operating System : Windows 10 or later; Linux or macOS
- Code editor : Any text editor or IDE, such as Visual Studio Code
- Programming language : Python 3
- Python database package : MySQL Connector/Python

# SOURCE CODE

## Source Code Summary

The project is implemented in a single Python file, `Car Showroom.py`, using MySQL Connector/Python to communicate with a local MySQL server. The `dbconnect()` function establishes the connection, and `setup()` creates the `carshowroom` database and it four tables: `Customers`, `Inventory`, `Staff`, and `Transactions`.

The program is organized into functions for adding, viewing, editing, and deleting records. `add_new()` handles new cars, staff, and customers; `edit_record()` checks that a selected record exists and limits updates to permitted columns; and `delete_record()` handles record deletion and quote cancellation. The shared `print_table()` function calculates column widths and displays aligned tables. Sample records can be added through `dummies()`.

Transaction functions manage quotes and sales. `txn()` checks the customer and car, calculates profit as sale price minus cost price, derives the quarter from the entered month, and updates inventory when recording a quote or finalized transaction. `finalise()` converts an existing quote to a sale, while `bill()` prints invoice details. Managers also have access to showroom statistics, quarterly reports, staff performance, customer history, and administrative options such as clearing table data.

At the end of the file, the program connects to MySQL, sets up the database, handles sample data and Manager creation, prompts for login, and runs the interactive menu. Transaction dates are checked for the expected format and a valid month. Staff passkeys are stored in plain text.

```python
'''==========CAR SHOWROOM DATABASE==========
Made by:
Joel Ephrem Cebin
Adhinath V. K.
Alan Anil'''
import math
import mysql.connector

db = None
crsr = None

def dbconnect(usr, passwd):
    global db, crsr
    try:
        db = mysql.connector.connect(host = 'localhost', user = usr, password = passwd)
    except mysql.connector.Error:
        return False
    crsr = db.cursor()
    return True

def setup():
    crsr.execute("create database if not exists carshowroom")
    crsr.execute("use carshowroom")
    crsr.execute("create table if not exists Customers (cust_id int primary key, name varchar (100) not null, phone varchar(15) not null, email varchar(100), address varchar(255))")
    crsr.execute("create table if not exists Inventory (car_id int primary key, make varchar (50) not null, model varchar(50) not null, year int not null, cost_price decimal(15, 2) not null, selling_price decimal(15, 2) not null, available_qty int default 0, reserved_qty int default 0, sold_qty int default 0)")
    crsr.execute("create table if not exists Staff (emp_id int primary key, name varchar(100) not null, passkey varchar(50), role varchar(20) not null, phone varchar(15) not null, salary decimal(10,2) not null, experience decimal(4, 2) not null)")
    crsr.execute("create table if not exists Transactions (txn_id int primary key, car_id int not null, cust_id int not null, emp_id int not null, sale_price decimal(15,2) not null, net_profit decimal(15,2) not null, status varchar(20) default 'Quote', txn_date date not null, quarter varchar(5) not null)")
    print("Table Creation/Repair Complete")


```

```
def dummies():
    try:
        crsr.execute("insert into Customers values (301, 'Naveen Nair', '9847123456', 'naveen.
        nair@gmail.com', 'Panampilly Nagar, Kochi'), (302, 'Fathima Rafeeq', '9447234567',
        'fathima.r@yahoo.com', 'Manacaud, Thiruvananthapuram'), (303, 'Rony Mathew',
        '9744345678', 'rony.mathew@outlook.com', 'Kanjikuzhy, Kottayam'), (304, 'Aisha
        Sulaiman', '9895456789', 'aisha.sulaiman@gmail.com', 'Palayam, Kozhikode'), (305,
        'Harikrishnan V.', '9496567890', 'hari.krishnan@gmail.com', 'Chembukkav, Thrissur')")
        crsr.execute("insert into Inventory values (201, 'Toyota', 'Innova Hycross ZX', 2025,
        2400000.00, 2750000.00, 3, 1, 2), (202, 'Hyundai', 'Creta SX(O)', 2025, 1500000.00,
        1720000.00, 4, 0, 3), (203, 'Mahindra', 'Thar Roxx AX7L', 2026, 1850000.00, 2100000.00,
        0, 1, 4), (204, 'Maruti Suzuki', 'Swift ZXi+', 2025, 750000.00, 880000.00, 5, 0, 4),
        (205, 'Tata', 'Nexon EV Empowered', 2026, 1450000.00, 1680000.00, 2, 0, 1)")
        crsr.execute("insert into Staff values (101, 'Mathew Thomas', 'admin123', 'Manager',
        '9846011223', 85000.00, 7.50), (102, 'Mohammed Asif', 'sales123', 'Sales',
        '9745122334', 38000.00, 3.00), (103, 'Sreekanth Pillai', 'sales456', 'Sales',
        '9633233445', 42000.00, 4.50), (104, 'Anju Varghese', 'sales789', 'Sales',
        '9447344556', 32000.00, 1.50)")
        crsr.execute("insert into Transactions values (401, 201, 301, 102, 2750000.00, 350000.
        00, 'Finalised', '2026-01-18', 'Q1'), (402, 204, 302, 102, 880000.00, 130000.00,
        'Finalised', '2026-04-12', 'Q2'), (403, 202, 301, 103, 1720000.00, 220000.00,
        'Finalised', '2026-07-25', 'Q3'), (404, 205, 304, 102, 1680000.00, 230000.00,
        'Finalised', '2026-10-05', 'Q4'), (405, 203, 303, 103, 2100000.00, 250000.00, 'Quote',
        '2026-11-14', 'Q4'), (406, 201, 301, 104, 2750000.00, 350000.00, 'Quote', '2026-12-02',
        'Q4')")
```

```
        db.commit()
        print("Dummy values have been inserted successfully")
        print("Dummy Admin Name: Mathew Thomas")
        print("Dummy Admin Password: admin123")
    except mysql.connector.Error:
        print("Unable to add Dummy Values - Primary Key Conflict")
        db.rollback()


def purge():
    if admin:
        print("WARNING: This operation will erase ALL DATA contained within the database.")
        print()
        conf = input("To confirm, type 'Yes, do as I say!': ")
        if conf == 'Yes, do as I say!':
            crsr.execute("truncate table Transactions")
            crsr.execute("truncate table Customers")
            crsr.execute("truncate table Inventory")
            crsr.execute("truncate table Staff")
            db.commit()
            print("Table Values Cleared")
        else:
            print("Deletion cancelled")
            return False
    else:
        print("This function requires elevated privileges")
        return False

```

```
65  def add_new(n):
66      '''
67      1 - New Car
68      2 - New Employee
69      3 - New Customer
70      '''
71      try:
72          if n == 1:
73              car_id = int(input("Enter Car ID: "))
74              if car_id <= 0:
75                  print("Failed: Car ID must be positive.")
76                  return False
77              make = input("Enter Make: ").strip()
78              model = input("Enter Model: ").strip()
79              year = int(input("Enter Year: "))
80              cost_price = float(input("Enter the Cost Price: "))
81              selling_price = float(input("Enter the Selling Price: "))
82              available_qty = int(input("Enter the Available Quantity: "))
83              if not make or not model or year <= 0 or available_qty < 0:
84                  print("Failed: Enter a make, model, positive year, and non-negative quantity.")
85                  return False
86              if not math.isfinite(cost_price) or not math.isfinite(selling_price) or cost_price < 0
                or selling_price <= 0:
87                  print("Failed: Prices must be finite; cost cannot be negative and selling price must
                    be positive.")
88                  return False
```

```
89              crsr.execute("INSERT INTO Inventory (car_id, make, model, year, cost_price,
                selling_price, available_qty) VALUES (%s,%s,%s,%s,%s,%s,%s)", (car_id, make, model,
                year, cost_price, selling_price, available_qty))
90              db.commit()
91              print("Car added successfully.")
92              return True
93
94          elif n == 2:
95              emp_id = int(input("Enter the Employee ID: "))
96              if emp_id <= 0:
97                  print("Failed: Employee ID must be positive.")
98                  return False
99              name = input("Enter the Name: ").strip()
100             passkey = input("Enter the Passkey: ")
101             role = input("Enter the Role: ").strip().title()
102             phone = input("Enter the Phone: ")
103             salary = float(input("Enter the Salary: "))
104             experience = float(input("Enter the Experience: "))
105             if not name or not phone or role not in ("Manager", "Sales"):
106                 print("Failed: Enter a name, phone, and a valid role (Manager or Sales).")
107                 return False
108             if not math.isfinite(salary) or not math.isfinite(experience) or salary < 0 or
                experience < 0:
109                 print("Failed: Salary and experience must be finite, non-negative values.")
110                 return False
111             crsr.execute("INSERT INTO Staff (emp_id, name, passkey, role, phone, salary, experience)
                VALUES (%s,%s,%s,%s,%s,%s,%s)", (emp_id, name, passkey, role, phone, salary, experience))
112             db.commit()
113             print("Employee added successfully.")
114             return True
115
```

```
        elif n == 3:
            cust_id = int(input("Enter the Customer ID: "))
            if cust_id <= 0:
                print("Failed: Customer ID must be positive.")
                return False
            name = input("Enter the Name: ").strip()
            phone = input("Enter the Phone: ").strip()
            email = input("Enter the Email: ")
            address = input("Enter the Address: ")
            if not name or not phone:
                print("Failed: Customer name and phone are required.")
                return False
            crsr.execute("INSERT INTO Customers (cust_id, name, phone, email, address) VALUES (%s,%s,
            %s,%s,%s)", (cust_id, name, phone, email, address))
            db.commit()
            print("Customer added successfully.")
            return True

        else:
            print("Failed: Invalid Option Selected.")
            return False

    except ValueError:
        print("Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).")
        return False
    except Exception as e:
        db.rollback()
        print("Failed to add record due to a database error:", e)
        return False


def edit_record(n):
    '''
    1 - Inventory
    2 - Staff
    3 - Customers
    '''
    tables = {
        1: ("Inventory", "car_id", ("make", "model", "year", "cost_price", "selling_price",
        "available_qty"), "Car"),
        2: ("Staff", "emp_id", ("name", "passkey", "role", "phone", "salary", "experience"),
        "Employee"),
        3: ("Customers", "cust_id", ("name", "phone", "email", "address"), "Customer"),
    }
```

```
    if n not in tables:
        print("Failed: Invalid Option Selected.")
        return False

    display_option = {1: 2, 2: 3, 3: 1}[n]
    if not disp(display_option):
        return False

    table, id_column, allowed_columns, record_name = tables[n]
    try:
        record_id = int(input("Enter " + record_name + " ID to edit: "))
        crsr.execute(
            "SELECT " + id_column + " FROM " + table + " WHERE " + id_column + " = %s",
            (record_id,),
        )
        if crsr.fetchone() is None:
            print(record_name + " ID not found.")
            return False
        col = input("Enter column to update (" + ", ".join(allowed_columns) + "): ").strip().lower()
        if col not in allowed_columns:
            print("Failed: Invalid column selected.")
            return False
        val = input("Enter new value for " + col + ": ")
        crsr.execute(
            "UPDATE " + table + " SET " + col + " = %s WHERE " + id_column + " = %s",
            (val, record_id),
        )
        if crsr.rowcount == 0:
            print("Failed: " + record_name + " ID not found or no changes were made.")
            return False
        db.commit()
        print(record_name + " record updated successfully.")
        return True

    except ValueError:
        print("Failed: Invalid input! Please enter a valid numeric ID.")
        return False
    except Exception as e:
        db.rollback()
        print("Failed to update record due to a database error:", e)
        return False


def select_open_quote(prompt):
    crsr.execute(
        "select Transactions.txn_id, Transactions.car_id, Inventory.make, Inventory.model, "
        "Customers.name, Staff.name, Transactions.sale_price, Transactions.txn_date "
        "from Transactions "
        "left join Inventory on Transactions.car_id = Inventory.car_id "
        "left join Customers on Transactions.cust_id = Customers.cust_id "
        "left join Staff on Transactions.emp_id = Staff.emp_id "
        "where Transactions.status = 'Quote' order by Transactions.txn_date"
    )
    quotes = crsr.fetchall()
    if not quotes:
        print("No open quotes available.")
        return None

    quote_ids = {quote[0] for quote in quotes}
    quote_rows = []
```

```
    for quote in quotes:
        car_name = " ".join(part for part in (quote[2], quote[3]) if part) or "Unknown car"
        quote_rows.append((
            quote[0],
            car_name,
            quote[4] or "Unknown customer",
            quote[5] or "Unknown employee",
            quote[6],
            quote[7],
        ))
    print_table(
        "OPEN QUOTES",
        ('Quote ID', 'Car', 'Customer', 'Employee', 'Sale Price', 'Date'),
        quote_rows,
    )
    try:
        quote_id = int(input(prompt))
    except ValueError:
        print("Invalid Input - Quote ID must be numerical.")
        return None
    if quote_id not in quote_ids:
        print("Quote ID not found in the open quote list.")
        return None
    return quote_id

def delete_record(n):
    '''
    1 - Inventory
    2 - Staff
    3 - Customers
    4 - Quotes (To be Added)
    '''
    try:
        if n == 1:
            record_id = int(input("Enter Car ID to delete: "))
            crsr.execute("DELETE FROM Inventory WHERE car_id = %s", (record_id,))
            if crsr.rowcount == 0:
                print("Failed: Car ID not found.")
                return False
            db.commit()
            print("Car deleted successfully.")
            return True
```

```
260         elif n == 2:
261             record_id = int(input("Enter Employee ID to delete: "))
262             crsr.execute("DELETE FROM Staff WHERE emp_id = %s", (record_id,))
263             if crsr.rowcount == 0:
264                 print("Failed: Employee ID not found.")
265                 return False
266             db.commit()
267             print("Employee deleted successfully.")
268             return True
269 
270         elif n == 3:
271             record_id = int(input("Enter Customer ID to delete: "))
272             crsr.execute("DELETE FROM Customers WHERE cust_id = %s", (record_id,))
273             if crsr.rowcount == 0:
274                 print("Failed: Customer ID not found.")
275                 return False
276             db.commit()
277             print("Customer deleted successfully.")
278             return True
279 
280         elif n == 4:
281             record_id = select_open_quote("Enter the Quote ID to cancel: ")
282             if record_id is None:
283                 return False
284             crsr.execute(
285                 "select car_id from Transactions where status = 'Quote' and txn_id = %s",
286                 (record_id,),
287             )
288             tup = crsr.fetchone()
289             if tup is None:
290                 print("Quote ID not found in the open quote list.")
291                 return False
292             car_id = tup[0]
293             crsr.execute("delete from Transactions where txn_id = %s", (record_id,))
294             if crsr.rowcount == 0:
295                 db.rollback()
296                 print("Quote was not deleted.")
297                 return False
298             crsr.execute("update Inventory set available_qty = available_qty + 1, reserved_qty =
                reserved_qty - 1 where car_id = %s and reserved_qty > 0", (car_id,))
299             if crsr.rowcount == 0:
300                 db.rollback()
301                 print("Failed to restore the reserved car quantity.")
302                 return False
303             db.commit()
304             print("Quote Deleted Successfully")
305             return True
306 
307         else:
308             print("Failed: Invalid Option Selected.")
309             return False
310 
311     except ValueError:
312         print("Failed: Invalid input! Please enter a valid numeric ID.")
313         return False
314     except Exception as e:
315         db.rollback()
316         print("Failed to delete record due to a database error:", e)
317         return False
318 
319 
```

```python
def auth(name, psky):
    global emp, admin
    emp = name.strip()
    crsr.execute("select role from Staff where name = %s and passkey = %s", (name, psky.strip()))
    posn = crsr.fetchone()
    if posn is not None:
        if posn[0] == 'Manager':
            print("Login Successful (Admin Mode)")
            admin = True
            return True
        else:
            print("Login Successful")
            admin = False
            return True
    else:
        print("User Not Found")
        admin = False
        return False

def bill(txn_id):
    crsr.execute("select * from Transactions where txn_id = %s", (txn_id,))
    data = crsr.fetchone()
    if data is None:
        print("Invalid Transaction ID")
        return False
    date = data[7]
    crsr.execute("select model from Inventory where car_id = %s", (data[1],))
    carname = crsr.fetchone()
    if carname is not None:
        carname = carname[0]
    else:
        print("Car not found")
        return False
    crsr.execute("select name from Customers where cust_id = %s", (data[2],))
    cust_name = crsr.fetchone()
    if cust_name is not None:
        cust_name = cust_name[0]
    else:
        print("Customer Name not found")
        return False
    crsr.execute("select name from Staff where emp_id = %s", (data[3],))
    emp_name = crsr.fetchone()
    if emp_name is not None:
        emp_name = emp_name[0]
    else:
        print("Employee Name not found")
        return False
    print("===============  INVOICE  ===============")
    print("DATE:", date)
    print("TRANSACTION ID:", txn_id)
    print("CAR:", carname)
    print("CUSTOMER:", cust_name)
    print("EMPLOYEE:", emp_name)
    print("PRICE:", data[4])
    print("STATUS:", data[6])
    print("-----------------------------------------")
    print()
    return True

```

```python
def txn():
    try:
        txn_id = int(input("Enter a Transaction ID: "))
        if txn_id <= 0:
            print("Transaction ID must be positive.")
            return False
    except ValueError:
        print("Invalid Input")
        return False
    crsr.execute("select txn_id from Transactions where txn_id = %s", (txn_id,))
    if crsr.fetchone() is not None:
        print("Transaction ID already in use")
        return False
    crsr.execute("select * from Customers")
    recs = crsr.fetchall()
    print_table("AVAILABLE CUSTOMERS", ('Customer ID', 'Name', 'Phone', 'Email', 'Address'), recs)
    print()
    try:
        cust_id = int(input("Enter the Customer ID: "))
        if cust_id <= 0:
            print("Customer ID must be positive.")
            return False
    except ValueError:
        print("Invalid Input")
        return False
    crsr.execute("select cust_id from Customers where cust_id = %s", (cust_id,))
    if crsr.fetchone() is None:
        print("Invalid Customer ID")
        return False
    crsr.execute("select * from Inventory")
    recs = crsr.fetchall()
    print_table(
        "AVAILABLE INVENTORY",
        ('Car ID', 'Make', 'Model', 'Year', 'Cost Price', 'Selling Price', 'Available', 'Reserved',
        'Sold'),
        recs,
    )
    print()
    try:
        car_id = int(input("Enter the Car ID: "))
        if car_id <= 0:
            print("Car ID must be positive.")
            return False
    except ValueError:
        print("Invalid Input")
        return False
```

```
424     crsr.execute("select emp_id from Staff where name = %s", (emp,))
425     emp_id = crsr.fetchone()
426     if emp_id is None:
427         print("Invalid Employee ID")
428         return False
429     emp_id = emp_id[0]
430     try:
431         sp = float(input("Enter the Sale Price: "))
432         if not math.isfinite(sp) or sp <= 0:
433             print("Sale Price must be a finite value greater than zero.")
434             return False
435     except ValueError:
436         print("Invalid Input")
437         return False
438     crsr.execute("select cost_price, available_qty from Inventory where car_id = %s", (car_id,))
439     cp = crsr.fetchone()
440     if cp is None:
441         print("Invalid Car ID")
442         return False
443     if cp[1] < 1:
444         print("The Entered Product is Out of Stock")
445         return False
446     cp = float(cp[0])
447     netprof = sp - cp
448     print("The net profit from this transaction will be", netprof)
449     status = input("Enter the Status (Quote/Finalised): ").strip().title()
450     if status not in ('Quote', 'Finalised'):
451         print("Invalid Status")
452         return False
453     txn_date = input("Enter the transaction date (YYYY-MM-DD): ")
454     split = txn_date.split('-')
455     if len(split) != 3:
456         print("Invalid Date Format Used")
457         return False
458     if not (split[0].isdigit() and split[1].isdigit() and split[2].isdigit()):
459         print("Invalid Date Format Used")
460         return False
461     if split[1] in ('01', '02', '03'):
462         q = "Q1"
463     elif split[1] in ('04', '05', '06'):
464         q = "Q2"
465     elif split[1] in ('07', '08', '09'):
466         q = "Q3"
467     elif split[1] in ('10', '11', '12'):
468         q = "Q4"
469     else:
470         print("Invalid Month Value")
471         return False
```

```
    try:
        if status == 'Finalised':
            crsr.execute("update Inventory set available_qty = available_qty - 1, sold_qty =
            sold_qty + 1 where car_id = %s and available_qty > 0", (car_id,))
        elif status == 'Quote':
            crsr.execute("update Inventory set available_qty = available_qty - 1, reserved_qty =
            reserved_qty + 1 where car_id = %s and available_qty > 0", (car_id,))
        if crsr.rowcount == 0:
            db.rollback()
            print("The car is no longer available in stock.")
            return False
        crsr.execute("insert into Transactions values (%s, %s, %s, %s, %s, %s, %s, %s, %s)",
        (txn_id, car_id, cust_id, emp_id, sp, netprof, status, txn_date, q))
        db.commit()
    except mysql.connector.Error as err:
        db.rollback()
        print("Database Error -", err)
        return False
    print(f"Transaction Recorded as {status} Successfully")
    print()
    prnt = input("Print bill? (y/n): ")
    if prnt.lower() == "y":
        status = bill(txn_id)
        if status == False:
            print("Unable to Print Bill")
    return True

def finalise(txn_id):
    crsr.execute("select * from Transactions where txn_id = %s", (txn_id,))
    txn = crsr.fetchone()
    if txn is None:
        print("Invalid Transaction ID")
        return False
    if txn[6] != "Quote":
        print("Only quote transactions can be finalised.")
        return False
    car_id = txn[1]
    try:
        crsr.execute("update Transactions set status = 'Finalised' where txn_id = %s and status =
        'Quote'", (txn_id,))
        if crsr.rowcount == 0:
            db.rollback()
            print("This transaction is no longer an open quote.")
            return False
        crsr.execute("update Inventory set sold_qty = sold_qty + 1, reserved_qty = reserved_qty - 1
        where car_id = %s and reserved_qty > 0", (car_id,))
        if crsr.rowcount == 0:
            db.rollback()
            print("Failed to find the reserved car quantity.")
            return False
        db.commit()
```

```
    except mysql.connector.Error as err:
        db.rollback()
        print("Update failed:", err)
        return False
    print("Transaction Finalised Successfully")
    print()
    inv = input("Do you want to print the bill? (y/n): ").lower()
    if inv == 'y':
        status = bill(txn_id)
        if status == False:
            print("Bill Printing Failed")
    return True

def print_table(title, headers, rows):
    rows = [[str(value) for value in row] for row in rows]
    widths = [
        max(len(str(header)), max((len(row[index]) for row in rows), default=0))
        for index, header in enumerate(headers)
    ]
    table_width = sum(widths) + 3 * (len(widths) - 1)

    def print_row(row):
        print(" | ".join(value.ljust(widths[index]) for index, value in enumerate(row)))

    print(title.center(table_width))
    print("=" * table_width)
    print_row([str(header) for header in headers])
    print("-+-".join("-" * width for width in widths))
    for row in row (variable) row: list[str]
        print_row(row)


def disp(n):
    '''
    1 - Customers
    2 - Inventory
    3 - Staff
    4 - Transactions
    '''
    if n == 1:
        crsr.execute("select * from Customers")
        data = crsr.fetchall()
        if not data:
            print("No Data in Table")
            print()
            return False
        print_table("CUSTOMERS", ('Customer ID', 'Name', 'Phone', 'Email', 'Address'), data)
        return True
    elif n == 2:
        crsr.execute("select * from Inventory")
        data = crsr.fetchall()
        if not data:
            print("No Data in Table")
            print()
            return False
        print_table(
            "INVENTORY",
            ('Car ID', 'Make', 'Model', 'Year', 'Cost Price', 'Selling Price', 'Available',
            'Reserved', 'Sold'),
            data,
        )
        return True
```

```
    elif n == 3:
        if admin:
            crsr.execute("select emp_id, name, role, phone, salary, experience from Staff")
            data = crsr.fetchall()
            if not data:
                print("No Data in Table")
                return False
            print_table("STAFF", ('Employee ID', 'Name', 'Role', 'Phone', 'Salary', 'Experience'),
            data)
            return True
        else:
            print("Access Denied: Table requires Elevated Privileges to View")
            return False
    elif n == 4:
        crsr.execute(
            "select Transactions.txn_id, Inventory.make, Inventory.model, Customers.name, Staff.
            name, "
            "Transactions.sale_price, Transactions.net_profit, Transactions.status, "
            "Transactions.txn_date, Transactions.quarter "
            "from Transactions "
            "left join Inventory on Transactions.car_id = Inventory.car_id "
            "left join Customers on Transactions.cust_id = Customers.cust_id "
            "left join Staff on Transactions.emp_id = Staff.emp_id"
        )
        data = crsr.fetchall()
        if not data:
            print("No Data in Table")
            print()
            return False
        display_data = []
        for row in data:
            car_name = " ".join(part for part in (row[1], row[2]) if part) or "Unknown car"
            display_data.append((
                row[0],
                car_name,
                row[3] or "Unknown customer",
                row[4] or "Unknown employee",
                row[5],
                row[6],
                row[7],
                row[8],
                row[9],
            ))
        print_table(
            "TRANSACTIONS",
            ('Txn ID', 'Car', 'Customer', 'Employee', 'Sale Price', 'Net Profit', 'Status', 'Date',
            'Quarter'),
            display_data,
        )
        return True
    else:
        print("Invalid Option")
        return False

```

```
630 def q_report():
631     if admin:
632         crsr.execute("select quarter, count(txn_id), sum(sale_price), sum(net_profit), avg
        (net_profit) from Transactions where status = 'Finalised' group by quarter order by quarter")
633         data = crsr.fetchall()
634         if data == []:
635             print("No data to show")
636             return False
637         print("========== QUARTERLY REPORT ==========")
638         print("--------------------------------------")
639         tsales, tprofit = 0, 0
640         for i in data:
641             print("Quarter:", i[0])
642             print("Transactions:", i[1])
643             print("Total Sale Price:", i[2])
644             print("Total Profit:", i[3])
645             print("Average Profit:", i[4])
646             print("--------------------------------------")
647             tsales += i[2]
648             tprofit += i[3]
649         print("Total Sales:", tsales)
650         print("Total Profit:", tprofit)
651         print("======================================")
652         return True
653     else:
654         print("This function requires elevated privileges.")
655         return False
656
657 def emp_stats():
658     if admin:
659         crsr.execute("select Staff.emp_id, Staff.name, Staff.experience, count(Transactions.txn_id),
        sum(Transactions.net_profit) from Staff, Transactions where Staff.emp_id = Transactions.
        emp_id and Transactions.status = 'Finalised' group by Staff.name, Staff.experience, Staff.
        emp_id order by count(Transactions.txn_id) desc")
660         data = crsr.fetchall()
661         if data == []:
662             print("No Data Available")
663             return False
664         print("========== STAFF SALES REPORT ==========")
665         print("----------------------------------------")
666         for i in data:
667             print("Employee ID:", i[0])
668             print("Name:", i[1])
669             print("Experience:", i[2])
670             print("Sales:", i[3])
671             print("Total Profit:", i[4])
672             print("----------------------------------------")
673         print("========================================")
674         return True
675     else:
676         print("This function requires elevated privileges.")
677         return False
678
```

```python
def cust_history():
    try:
        cust_id = int(input("Enter the Customer ID: "))
    except ValueError:
        print("Invalid Input")
        return False

    crsr.execute("select name from Customers where cust_id = %s", (cust_id,))
    cust_name = crsr.fetchone()

    if cust_name is None:
        print("Invalid Customer ID")
        return False

    print("\n========== PURCHASE HISTORY ==========")
    print("CUSTOMER: ", cust_name[0])
    print("--------------------------------------")

    crsr.execute("select Transactions.txn_id, Transactions.txn_date, Inventory.make, Inventory.
    model, Transactions.sale_price, Transactions.status from Transactions, Inventory where
    Transactions.car_id = Inventory.car_id and Transactions.cust_id = %s order by Transactions.
    txn_date desc", (cust_id,))
    recs = crsr.fetchall()

    if recs == []:
        print("No purchase history found for this customer.")
    else:
        for i in recs:
            print("Txn ID:", i[0])
            print("Date:", i[1])
            print("Make:", i[2])
            print("Model:", i[3])
            print("Price:", i[4])
            print("Status:", i[5])
            print("--------------------------------------")
    print("======================================")
    return True

def show_statistics():
    try:
        print("\n========== SHOWROOM STATISTICS ==========")

        crsr.execute("SELECT SUM(available_qty) FROM Inventory")
        total_cars = crsr.fetchone()[0]
        if total_cars == None:
            total_cars = 0
        print("Total Cars in Stock:", total_cars)

        crsr.execute("SELECT SUM(cost_price * available_qty) FROM Inventory")
        inventory_value = crsr.fetchone()[0]
        if inventory_value == None:
            inventory_value = 0
        print("Total Inventory Value (Cost): ₹" + str(inventory_value))

        crsr.execute("SELECT COUNT(cust_id) FROM Customers")
        total_customers = crsr.fetchone()[0]
        if total_customers == None:
            total_customers = 0
        print("Total Registered Customers:", total_customers)

```

```
        crsr.execute("SELECT COUNT(emp_id) FROM Staff")
        total_staff = crsr.fetchone()[0]
        if total_staff == None:
            total_staff = 0
        print("Total Staff Members:", total_staff)

        print("========================================\n")
        return True

    except Exception as e:
        print("Failed to retrieve statistics due to a database error:", e)
        return False

def _run_program():
    while True:
        if dbconnect(input("Enter your MySQL Username: "), input("Enter your MySQL Password: ")):
            print("Database Connection Successful")
            break
        else:
            print("Database Connection Failed")
        print()

    setup()
    opt = input("Do you want to add dummy values into the database for testing? (y/n): ")
    if opt.lower() == 'y':
        dummies()
    else:
        print("Dummy values can be added later if you change your mind.")
    print()
    crsr.execute("select count(*) from Staff where Role = 'Manager'")
    data = crsr.fetchone()
    if data[0] == 0:
        while True:
            print("No Manager Data is Available in the Database.")
            print("Please add a Manager account to continue")
            if add_new(2):
                crsr.execute("select count(*) from Staff where Role = 'Manager'")
                data = crsr.fetchone()
                if data[0] != 0:
                    print()
                    break

    while True:
        if auth(input("Enter your Name: "), input("Enter your Passkey: ")):
            print()
            break
        print()

    while True:
        print()
        print("==========CAR SHOWROOM DATABASE==========")
        print("|        PLEASE SELECT AN OPTION        |")
        print("|        [1] Customer Details           |")
        print("|        [2] Inventory Management       |")
        print("|        [3] Transactions               |")
        print("|        [4] Exit                       |")
        if admin:
            print("|        [5] Staff Details              |")
            print("|        [6] Statistics                 |")
            print("|        [7] Administrator Menu         |")
        print("+---------------------------------------+")
        opt = input("       Select an option: ").strip()
```

```
799         if opt == '1':
800             while True:
801                 print()
802                 print("[1] Add New Customer")
803                 print("[2] View All Customers")
804                 print("[3] Update Customer Details")
805                 print("[4] Delete Customer Details")
806                 print("[5] Back")
807                 sub_opt = input('Select an Option: ').strip()
808                 if sub_opt == '1':
809                     add_new(3)
810                 elif sub_opt == '2':
811                     disp(1)
812                 elif sub_opt == '3':
813                     edit_record(3)
814                 elif sub_opt == '4':
815                     delete_record(3)
816                 elif sub_opt == '5':
817                     break
818                 else:
819                     print("Invalid Option")
820 
821         elif opt == '2':
822             while True:
823                 print()
824                 print("[1] Add a New Car")
825                 print("[2] View Inventory")
826                 print("[3] Update Car Details")
827                 print("[4] Delete a Car")
828                 print("[5] Back")
829                 sub_opt = input('Select an Option: ').strip()
830                 if sub_opt == '1':
831                     add_new(1)
832                 elif sub_opt == '2':
833                     disp(2)
834                 elif sub_opt == '3':
835                     edit_record(1)
836                 elif sub_opt == '4':
837                     delete_record(1)
838                 elif sub_opt == '5':
839                     break
840                 else:
841                     print("Invalid Option")
842 
843         elif opt == '3':
844             while True:
845                 print()
846                 print("[1] Create a New Quote/Transaction")
847                 print("[2] Finalise a Quote")
848                 print("[3] Cancel a Reservation")
849                 print("[4] View/Reprint an Invoice")
850                 print("[5] Show All Transactions")
851                 print("[6] Back")
852                 sub_opt = input('Select an Option: ').strip()
853                 if sub_opt == '1':
854                     txn()
855                 elif sub_opt == '2':
856                     quote_id = select_open_quote("Enter the Quote ID to finalise: ")
857                     if quote_id is not None:
858                         finalise(quote_id)
859                 elif sub_opt == '3':
860                     delete_record(4)
```

```
            elif sub_opt == '4':
                try:
                    bill(int(input("Enter the Transaction ID: ")))
                except ValueError:
                    print("ERROR: Transaction ID must be Numerical")
            elif sub_opt == '5':
                disp(4)
            elif sub_opt == '6':
                break
            else:
                print("Invalid Option")

    elif opt == '4':
        print()
        print("Goodbye.")
        break

    elif opt == '5':
        if admin:
            while True:
                print()
                print("[1] New Staff")
                print("[2] View all Staff")
                print("[3] Update Staff Details")
                print("[4] Delete a Staff Member")
                print("[5] Back")
                sub_opt = input('Select an Option: ').strip()
                if sub_opt == '1':
                    add_new(2)
                elif sub_opt == '2':
                    disp(3)
                elif sub_opt == '3':
                    edit_record(2)
                elif sub_opt == '4':
                    delete_record(2)
                elif sub_opt == '5':
                    break
                else:
                    print("Invalid Option")
        else:
            print("ERROR: Not Enough Permissions")
```

```
903         elif opt == '6':
904             if admin:
905                 while True:
906                     print()
907                     print("[1] Showroom Overview")
908                     print("[2] Quarterly Report")
909                     print("[3] Staff Performance")
910                     print("[4] Customer History")
911                     print("[5] Back")
912                     sub_opt = input('Select an Option: ').strip()
913                     if sub_opt == '1':
914                         show_statistics()
915                     elif sub_opt == '2':
916                         q_report()
917                     elif sub_opt == '3':
918                         emp_stats()
919                     elif sub_opt == '4':
920                         cust_history()
921                     elif sub_opt == '5':
922                         break
923                     else:
924                         print("Invalid Option")
925             else:
926                 print("ERROR: Not Enough Permissions")
927         elif opt == '7':
928             if admin:
929                 while True:
930                     print()
931                     print("[1] WIPE ALL DATA")
932                     print("[2] Insert Dummy Values")
933                     print("[3] Repair Tables")
934                     print("[4] Back")
935                     sub_opt = input('Select an Option: ').strip()
936                     if sub_opt == '1':
937                         purge()
938                     elif sub_opt == '2':
939                         dummies()
940                     elif sub_opt == '3':
941                         setup()
942                     elif sub_opt == '4':
943                         break
944                     else:
945                         print("Invalid Option")
946             else:
947                 print("ERROR: Not Enough Permissions")
948         else:
949             print("Invalid Option")
950
951 def main():
952     try:
953         _run_program()
954     finally:
955         try:
956             if crsr is not None:
957                 crsr.close()
958         finally:
959             if db is not None:
960                 db.close()
961
962
963 if __name__ == "__main__":
964     main()
965
966
```

# OUTPUT

**Execute the program and entered correct credentials to connect DB and Manager Role.**

```
PS C:\Users\ACER LAP\Desktop\Class12_Project\12th-Computer-Project> python "Car Showroom.py"
Enter your MySQL Username: root
Enter your MySQL Password: @Root$123
Database Connection Successful
Table Creation/Repair Complete
Do you want to add dummy values into the database for testing? (y/n): n
Dummy values can be added later if you change your mind.

Enter your Name: Suresh Kumar
Enter your Passkey: admin123
Login Successful (Admin Mode)
```

**Car Showroom DB Options are Displayed in the console and User can choose the options**

```
==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option:
```

**Option 1: Customer Details – Prompt another options Add/View/Update/Delete Customers or go back to Main menu.**

```
       Select an option: 1

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

# Valid Combinations for Customers:

**Option 2: View All Customers**

```
Select an Option: 2
                                        CUSTOMERS
=============================================================================================
Customer ID | Name           | Phone      | Email                    | Address
------------+----------------+------------+--------------------------+-----------------------------
301         | Naveen Nair    | 9847123456 | naveen.nair@gmail.com    | Panampilly Nagar, Kochi
302         | Fathima Rafeeq | 9447234567 | fathima.r@yahoo.com      | Manacaud, Thiruvananthapuram
303         | Rony Mathew    | 9744345678 | rony.mathew@outlook.com  | Kanjikuzhy, Kottayam
304         | Fathima Baker  | 9895456789 | aisha.sulaiman@gmail.com | Palayam, Kozhikode
305         | Remya Joseph   | 5657585945 | remya.joseph@test.com    | Muvattupuzha, Ernakulam

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 1: Add New Customer**

```
Select an Option: 1
Enter the Customer ID: 306
Enter the Name: Sony Mon Thomas
Enter the Phone: 3456789098
Enter the Email: sony.thomas@test.com
Enter the Address: Thodupuzha, Idukki
Customer added successfully.

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 2: View All Customers – List the newly added customer**

```
Select an Option: 2
                                        CUSTOMERS
=============================================================================================
Customer ID | Name           | Phone      | Email                    | Address
------------+----------------+------------+--------------------------+-----------------------------
301         | Naveen Nair    | 9847123456 | naveen.nair@gmail.com    | Panampilly Nagar, Kochi
302         | Fathima Rafeeq | 9447234567 | fathima.r@yahoo.com      | Manacaud, Thiruvananthapuram
303         | Rony Mathew    | 9744345678 | rony.mathew@outlook.com  | Kanjikuzhy, Kottayam
304         | Fathima Baker  | 9895456789 | aisha.sulaiman@gmail.com | Palayam, Kozhikode
305         | Remya Joseph   | 5657585945 | remya.joseph@test.com    | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas| 3456789098 | sony.thomas@test.com     | Thodupuzha, Idukki

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 3: Update Customer Details – Using the existing Customer ID**

```
Select an Option: 3
                                                CUSTOMERS
=============================================================================================================
Customer ID | Name             | Phone      | Email                     | Address
------------+------------------+------------+---------------------------+-----------------------------
301         | Naveen Nair      | 9847123456 | naveen.nair@gmail.com     | Panampilly Nagar, Kochi
302         | Fathima Rafeeq   | 9447234567 | fathima.r@yahoo.com       | Manacaud, Thiruvananthapuram
303         | Rony Mathew      | 9744345678 | rony.mathew@outlook.com   | Kanjikuzhy, Kottayam
304         | Fathima Baker    | 9895456789 | aisha.sulaiman@gmail.com  | Palayam, Kozhikode
305         | Remya Joseph     | 5657585945 | remya.joseph@test.com     | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas  | 3456789098 | sony.thomas@test.com      | Thodupuzha, Idukki
Enter Customer ID to edit: 305
Enter column to update (name, phone, email, address): name
Enter new value for name: Reetha Johny
Customer record updated successfully.

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 2: View All Customers – We can view the Customer ID 305's Name is changed to 'Reetha Johny'**

```
Select an Option: 2
                                                CUSTOMERS
=============================================================================================================
Customer ID | Name             | Phone      | Email                     | Address
------------+------------------+------------+---------------------------+-----------------------------
301         | Naveen Nair      | 9847123456 | naveen.nair@gmail.com     | Panampilly Nagar, Kochi
302         | Fathima Rafeeq   | 9447234567 | fathima.r@yahoo.com       | Manacaud, Thiruvananthapuram
303         | Rony Mathew      | 9744345678 | rony.mathew@outlook.com   | Kanjikuzhy, Kottayam
304         | Fathima Baker    | 9895456789 | aisha.sulaiman@gmail.com  | Palayam, Kozhikode
305         | Reetha Johny     | 5657585945 | remya.joseph@test.com     | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas  | 3456789098 | sony.thomas@test.com      | Thodupuzha, Idukki

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 4: Delete Customer Details – using the Customer ID**

```
Select an Option: 4
Enter Customer ID to delete: 303
Customer deleted successfully.

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 2: View All Customers – We can view the Customer ID 303 Deleted Successfully**

```
Select an Option: 2
                                        CUSTOMERS
=============================================================================================================
Customer ID | Name             | Phone      | Email                     | Address
------------+------------------+------------+---------------------------+-----------------------------
301         | Naveen Nair      | 9847123456 | naveen.nair@gmail.com     | Panampilly Nagar, Kochi
302         | Fathima Rafeeq   | 9447234567 | fathima.r@yahoo.com       | Manacaud, Thiruvananthapuram
304         | Fathima Baker    | 9895456789 | aisha.sulaiman@gmail.com  | Palayam, Kozhikode
305         | Reetha Johny     | 5657585945 | remya.joseph@test.com     | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas  | 3456789098 | sony.thomas@test.com      | Thodupuzha, Idukki

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

# Invalid Combinations for Customers and its validations:

**Option 1: Add New Customer – Customer ID does not allow Characters alone, Alphanumeric characters, Special Characters**

```
Select an Option: 1
Enter the Customer ID: rty
Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option: 1
Enter the Customer ID: 12rt
Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option: 1
Enter the Customer ID: @#$
Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).
```

**Option 3: Update Customer Details – Enter an invalid Customer ID**

```
Select an Option: 3
                                        CUSTOMERS
=============================================================================================================
Customer ID | Name             | Phone      | Email                     | Address
------------+------------------+------------+---------------------------+-----------------------------
301         | Naveen Nair      | 9847123456 | naveen.nair@gmail.com     | Panampilly Nagar, Kochi
302         | Fathima Rafeeq   | 9447234567 | fathima.r@yahoo.com       | Manacaud, Thiruvananthapuram
304         | Fathima Baker    | 9895456789 | aisha.sulaiman@gmail.com  | Palayam, Kozhikode
305         | Reetha Johny     | 5657585945 | remya.joseph@test.com     | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas  | 3456789098 | sony.thomas@test.com      | Thodupuzha, Idukki
Enter Customer ID to edit: 567
Customer ID not found.

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 3: Update Customer Details – Enter a valid Customer ID and misspell the column name, throws error message**

```
Select an Option: 3
                                        CUSTOMERS
=============================================================================================================
Customer ID | Name             | Phone      | Email                     | Address
------------+------------------+------------+---------------------------+-----------------------------
301         | Naveen Nair      | 9847123456 | naveen.nair@gmail.com     | Panampilly Nagar, Kochi
302         | Fathima Rafeeq   | 9447234567 | fathima.r@yahoo.com       | Manacaud, Thiruvananthapuram
304         | Fathima Baker    | 9895456789 | aisha.sulaiman@gmail.com  | Palayam, Kozhikode
305         | Reetha Johny     | 5657585945 | remya.joseph@test.com     | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas  | 3456789098 | sony.thomas@test.com      | Thodupuzha, Idukki
Enter Customer ID to edit: 305
Enter column to update (name, phone, email, address): naame
Failed: Invalid column selected.

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 4: Delete Customer Details – Enter an invalid Customer ID, throws error message**

```
Select an Option: 4
Enter Customer ID to delete: 708
Failed: Customer ID not found.

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Select an invalid option throws error message**

```
Select an Option: 6
Invalid Option

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option:
```

**Option 5: Back option will navigate back to main menu**

```
Select an Option: 5

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option:
```

**Option 2: Inventory Management– Prompt another options Add/View/Update/Delete from Inventory Management table or go back to Main menu**

```
       Select an option: 2

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

# Valid Combinations for Inventory Management

**Option 2: View Inventory**

```
Select an Option: 2
                                                    INVENTORY
=============================================================================================================
Car ID | Make          | Model              | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+--------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX  | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)        | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L     | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+         | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
205    | Tata          | Nexon EV Empowered | 2026 | 1450000.00 | 1680000.00    | 1         | 0        | 2
206    | Honda         | City               | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza             | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 1: Add a New Car**

```
Select an Option: 1
Enter Car ID: 208
Enter Make: Kia
Enter Model: Carens Clavis
Enter Year: 2026
Enter the Cost Price: 1100000
Enter the Selling Price: 1138900
Enter the Available Quantity: 50
Car added successfully.

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 2: View Inventory – List the newly added car**

```
Select an Option: 2
                                                    INVENTORY
=============================================================================================================
Car ID | Make          | Model               | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+---------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX   | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)         | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L      | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+          | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
205    | Tata          | Nexon EV Empowered  | 2026 | 1450000.00 | 1680000.00    | 1         | 0        | 2
206    | Honda         | City                | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza              | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis       | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 3: Update Car Details – Updated existing car details using Car ID**

```
Select an Option: 3
                                                    INVENTORY
=============================================================================================================
Car ID | Make          | Model               | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+---------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX   | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)         | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L      | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+          | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
205    | Tata          | Nexon EV Empowered  | 2026 | 1450000.00 | 1680000.00    | 1         | 0        | 2
206    | Honda         | City                | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza              | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis       | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0
Enter Car ID to edit: 205
Enter column to update (make, model, year, cost_price, selling_price, available_qty): model
Enter new value for model: Nexon EV Empowered Plus A 45
Car record updated successfully.

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 2: View Inventory – will display the updated car details**

```
Select an Option: 2
                                                    INVENTORY
=====================================================================================================================
Car ID | Make          | Model                       | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+-----------------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX           | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)                 | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L              | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+                  | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
205    | Tata          | Nexon EV Empowered Plus A 45 | 2026 | 1450000.00 | 1680000.00    | 1         | 0        | 2
206    | Honda         | City                        | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza                      | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis               | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 4: Delete a Car – using Car ID**

```
Enter Car ID to delete: 205
Car deleted successfully.

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 2: View Inventory – will display the deleted car details**

```
Select an Option: 2
                                                    INVENTORY
=================================================================================================================
Car ID | Make          | Model             | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+-------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)       | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L    | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+        | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
206    | Honda         | City              | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza            | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis     | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

# Invalid Combinations - Inventory Management and its validations:

**Option 1: Add New Car – Car ID does not allow Characters alone, Alphanumeric characters, Special Characters**

```
Select an Option: 1
Enter Car ID: tyyy
Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option: 1
Enter Car ID: 1TTT
Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option: 1
Enter Car ID: @#$%
Failed: Invalid input! Please enter the correct data type (e.g., numbers for IDs).

[1] Add a New Car
```

### Option 3: Update Car Details– Enter a valid Car ID and misspell the column name, throws error message

```
Select an Option: 3
                                                INVENTORY
=============================================================================================================
Car ID | Make          | Model             | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+-------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)       | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L    | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+        | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
206    | Honda         | City              | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza            | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis     | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0
Enter Car ID to edit: 208
Enter column to update (make, model, year, cost_price, selling_price, available_qty): tear
Failed: Invalid column selected.

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

### Option 3: Update Car Details – Enter an invalid Car ID

```
Select an Option: 3
                                                INVENTORY
=============================================================================================================
Car ID | Make          | Model             | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+-------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)       | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L    | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+        | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
206    | Honda         | City              | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza            | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis     | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0
Enter Car ID to edit: 307
Car ID not found.

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

### Option 4: Delete a Car – Enter an invalid Car ID, throws error message

```
Select an Option: 4
Enter Car ID to delete: 709
Failed: Car ID not found.

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

### Select an invalid option throws error message

```
Select an Option: 7
Invalid Option

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

**Option 5: Back option will navigate back to main menu**

```
Select an Option: 5

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option:
```

**Option 3: Transactions - Prompt another options Add/View/Update/Delete from Transactions table or go back to Main menu**

```
       Select an option: 3

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

# Valid Combinations for Inventory Management

**Option 5: Show All Transactions**

```
Select an Option: 5
                                                    TRANSACTIONS
=================================================================================================================================================
Txn ID | Car                      | Customer         | Employee         | Sale Price | Net Profit | Status    | Date       | Quarter
-------+--------------------------+------------------+------------------+------------+------------+-----------+------------+--------
401    | Toyota Innova Hycross ZX | Naveen Nair      | Mohammed Asif    | 2750000.00 | 350000.00  | Finalised | 2026-01-18 | Q1
402    | Maruti Suzuki Swift ZXi+ | Fathima Rafeeq   | Mohammed Asif    | 880000.00  | 130000.00  | Finalised | 2026-04-12 | Q2
403    | Hyundai Creta SX(O)      | Naveen Nair      | Sreekanth Pillai | 1720000.00 | 220000.00  | Finalised | 2026-07-25 | Q3
404    | Unknown car              | Fathima Baker    | Mohammed Asif    | 1680000.00 | 230000.00  | Finalised | 2026-10-05 | Q4
405    | Mahindra Thar Roxx AX7L  | Unknown customer | Sreekanth Pillai | 2100000.00 | 250000.00  | Quote     | 2026-11-14 | Q4
406    | Toyota Innova Hycross ZX | Naveen Nair      | Anju Varghese    | 2750000.00 | 350000.00  | Quote     | 2026-12-02 | Q4
407    | Unknown car              | Naveen Nair      | Suresh Kumar     | 1500000.00 | 50000.00   | Finalised | 2026-09-27 | Q3
408    | Hyundai Creta SX(O)      | Fathima Baker    | Suresh Kumar     | 1710000.00 | 210000.00  | Finalised | 2026-09-27 | Q3
409    | Toyota Innova Hycross ZX | Reetha Johny     | Suresh Kumar     | 2700000.00 | 300000.00  | Finalised | 2026-09-27 | Q3
410    | Maruti Suzuki Brezza     | Reetha Johny     | Suresh Kumar     | 1000000.00 | 260000.00  | Quote     | 2026-09-27 | Q3

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

**Option 1: Create a New Quote/Transaction – Prompt to Enter a Transaction ID, enter a valid Transaction ID, it will display Available Customer Details in the DB, prompt to Enter the Customer ID. Once entered valid Customer ID from list, it will display available Car details in the DB, prompt to Enter the Car ID. Once entered valid Car ID from the list, it will ask for further details to 'Create a New Quote/Transaction'**

### a) Creating a Quote:

```
Select an Option: 1
Enter a Transaction ID: 412
                                        AVAILABLE CUSTOMERS
=============================================================================================================
Customer ID | Name            | Phone      | Email                    | Address
------------+-----------------+------------+--------------------------+-----------------------------
301         | Naveen Nair     | 9847123456 | naveen.nair@gmail.com    | Panampilly Nagar, Kochi
302         | Fathima Rafeeq  | 9447234567 | fathima.r@yahoo.com      | Manacaud, Thiruvananthapuram
304         | Fathima Baker   | 9895456789 | aisha.sulaiman@gmail.com | Palayam, Kozhikode
305         | Reetha Johny    | 5657585945 | remya.joseph@test.com    | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas | 3456789098 | sony.thomas@test.com     | Thodupuzha, Idukki

Enter the Customer ID: 301
                                          AVAILABLE INVENTORY
=============================================================================================================
Car ID | Make          | Model             | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+-------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)       | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L    | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+        | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
206    | Honda         | City              | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza            | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis     | 2026 | 1100000.00 | 1138900.00    | 50        | 0        | 0

Enter the Car ID: 208
Enter the Sale Price: 1138900
The net profit from this transaction will be 38900.0
Enter the Status (Quote/Finalised): Quote
Enter the transaction date (YYYY-MM-DD): 2026-09-30
```

**After creating a Quote, system will prompt to print bill, Choose y**

```
Print bill? (y/n): y
===============  INVOICE  ===============
DATE: 2026-09-30
TRANSACTION ID: 412
CAR: Carens Clavis
CUSTOMER: Naveen Nair
EMPLOYEE: Suresh Kumar
PRICE: 1138900.00
STATUS: Quote
-----------------------------------------


[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

## b) Creating a New Transaction (Enter the Status (Quote/Finalised): Finalised)

```
Select an Option: 1
Enter a Transaction ID: 413
                                        AVAILABLE CUSTOMERS
=============================================================================================
Customer ID | Name            | Phone      | Email                   | Address
------------+-----------------+------------+-------------------------+---------------------------
301         | Naveen Nair     | 9847123456 | naveen.nair@gmail.com   | Panampilly Nagar, Kochi
302         | Fathima Rafeeq  | 9447234567 | fathima.r@yahoo.com     | Manacaud, Thiruvananthapuram
304         | Fathima Baker   | 9895456789 | aisha.sulaiman@gmail.com | Palayam, Kozhikode
305         | Reetha Johny    | 5657585945 | remya.joseph@test.com   | Muvattupuzha, Ernakulam
306         | Sony Mon Thomas | 3456789098 | sony.thomas@test.com    | Thodupuzha, Idukki

Enter the Customer ID: 302
                                        AVAILABLE INVENTORY
=============================================================================================================
Car ID | Make          | Model              | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+--------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX  | 2025 | 2400000.00 | 2750000.00    | 2         | 1        | 3
202    | Hyundai       | Creta SX(O)        | 2025 | 1500000.00 | 1720000.00    | 3         | 0        | 4
203    | Mahindra      | Thar Roxx AX7L     | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+         | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
206    | Honda         | City               | 2026 | 1700000.00 | 2000000.00    | 10        | 0        | 0
207    | Maruti Suzuki | Brezza             | 2026 | 740000.00  | 1000000.00    | 14        | 1        | 0
208    | Kia           | Carens Clavis      | 2026 | 1100000.00 | 1138900.00    | 49        | 1        | 0

Enter the Car ID: 204
Enter the Sale Price: 800000
The net profit from this transaction will be 50000.0
Enter the Status (Quote/Finalised): Finalised
Enter the transaction date (YYYY-MM-DD): 2026-09-30
```

## After creating a Quote, system will prompt to print bill, Choose n

```
Print bill? (y/n): n

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

## Option 5: Show All Transactions – Newly added Transactions are listed in the table (Txn ID 412,413)

```
Select an Option: 5
                                                    TRANSACTIONS
=============================================================================================================================================
Txn ID | Car                      | Customer         | Employee         | Sale Price | Net Profit | Status    | Date       | Quarter
-------+--------------------------+------------------+------------------+------------+------------+-----------+------------+--------
401    | Toyota Innova Hycross ZX | Naveen Nair      | Mohammed Asif    | 2750000.00 | 350000.00  | Finalised | 2026-01-18 | Q1
402    | Maruti Suzuki Swift ZXi+ | Fathima Rafeeq   | Mohammed Asif    | 880000.00  | 130000.00  | Finalised | 2026-04-12 | Q2
403    | Hyundai Creta SX(O)      | Naveen Nair      | Sreekanth Pillai | 1720000.00 | 220000.00  | Finalised | 2026-07-25 | Q3
404    | Unknown car              | Fathima Baker    | Mohammed Asif    | 1680000.00 | 230000.00  | Finalised | 2026-10-05 | Q4
405    | Mahindra Thar Roxx AX7L  | Unknown customer | Sreekanth Pillai | 2100000.00 | 250000.00  | Quote     | 2026-11-14 | Q4
406    | Toyota Innova Hycross ZX | Naveen Nair      | Anju Varghese    | 2750000.00 | 350000.00  | Quote     | 2026-12-02 | Q4
407    | Unknown car              | Naveen Nair      | Suresh Kumar     | 1500000.00 | 50000.00   | Finalised | 2026-09-27 | Q3
408    | Hyundai Creta SX(O)      | Fathima Baker    | Suresh Kumar     | 1710000.00 | 210000.00  | Finalised | 2026-09-27 | Q3
409    | Toyota Innova Hycross ZX | Reetha Johny     | Suresh Kumar     | 2700000.00 | 300000.00  | Finalised | 2026-09-27 | Q3
410    | Maruti Suzuki Brezza     | Reetha Johny     | Suresh Kumar     | 1000000.00 | 260000.00  | Quote     | 2026-09-27 | Q3
412    | Kia Carens Clavis        | Naveen Nair      | Suresh Kumar     | 1138900.00 | 38900.00   | Quote     | 2026-09-30 | Q3
413    | Maruti Suzuki Swift ZXi+ | Fathima Rafeeq   | Suresh Kumar     | 800000.00  | 50000.00   | Finalised | 2026-09-30 | Q3

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

**Option 2: Finalise a Quote – This will list all the Open Quotes, and prompt to Enter the Quote ID to finalise. After entering valid Transaction ID will provide the message 'Transaction Finalised Successfully.' Also prompt to print the bill, entered 'y,' and printed the bill in console**

```
Select an Option: 2
                                        OPEN QUOTES
=============================================================================================
Quote ID | Car                      | Customer         | Employee         | Sale Price | Date
---------+--------------------------+------------------+------------------+------------+-----------
410      | Maruti Suzuki Brezza     | Reetha Johny     | Suresh Kumar     | 1000000.00 | 2026-09-27
412      | Kia Carens Clavis        | Naveen Nair      | Suresh Kumar     | 1138900.00 | 2026-09-30
405      | Mahindra Thar Roxx AX7L  | Unknown customer | Sreekanth Pillai | 2100000.00 | 2026-11-14
406      | Toyota Innova Hycross ZX | Naveen Nair      | Anju Varghese    | 2750000.00 | 2026-12-02
Enter the Quote ID to finalise: 412
Transaction Finalised Successfully

Do you want to print the bill? (y/n): y
===============  INVOICE  ===============
DATE: 2026-09-30
TRANSACTION ID: 412
CAR: Carens Clavis
CUSTOMER: Naveen Nair
EMPLOYEE: Suresh Kumar
PRICE: 1138900.00
STATUS: Finalised
-----------------------------------------


[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

**Option 3: Cancel a Reservation - This will list all the Open Quotes, and prompt to Enter the Quote ID to 'Cancel a Reservation'. After entering valid Transaction ID will provide the message 'Quote Deleted Successfully'**

```
Select an Option: 3
                                        OPEN QUOTES
=============================================================================================
Quote ID | Car                      | Customer         | Employee         | Sale Price | Date
---------+--------------------------+------------------+------------------+------------+----------
410      | Maruti Suzuki Brezza     | Reetha Johny     | Suresh Kumar     | 1000000.00 | 2026-09-27
405      | Mahindra Thar Roxx AX7L  | Unknown customer | Sreekanth Pillai | 2100000.00 | 2026-11-14
406      | Toyota Innova Hycross ZX | Naveen Nair      | Anju Varghese    | 2750000.00 | 2026-12-02
Enter the Quote ID to cancel: 406
Quote Deleted Successfully

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

**Option 4: View/Reprint an Invoice**

**a) Finalised Transaction**

```
Select an Option: 4
Enter the Transaction ID: 401
===============  INVOICE  ===============
DATE: 2026-01-18
TRANSACTION ID: 401
CAR: Innova Hycross ZX
CUSTOMER: Naveen Nair
EMPLOYEE: Mohammed Asif
PRICE: 2750000.00
STATUS: Finalised
-----------------------------------------


[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

### b) Quote

```
Select an Option: 4
Enter the Transaction ID: 410
===============  INVOICE  ===============
DATE: 2026-09-27
TRANSACTION ID: 410
CAR: Brezza
CUSTOMER: Reetha Johny
EMPLOYEE: Suresh Kumar
PRICE: 1000000.00
STATUS: Quote
-----------------------------------------


[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

### Option 5: Show All Transactions – after all options in Transaction Table

```
Select an Option: 5
                                                    TRANSACTIONS
=================================================================================================================================================
Txn ID | Car                      | Customer         | Employee         | Sale Price | Net Profit | Status    | Date       | Quarter
-------+--------------------------+------------------+------------------+------------+------------+-----------+------------+--------
401    | Toyota Innova Hycross ZX | Naveen Nair      | Mohammed Asif    | 2750000.00 | 350000.00  | Finalised | 2026-01-18 | Q1
402    | Maruti Suzuki Swift ZXi+ | Fathima Rafeeq   | Mohammed Asif    | 880000.00  | 130000.00  | Finalised | 2026-04-12 | Q2
403    | Hyundai Creta SX(O)      | Naveen Nair      | Sreekanth Pillai | 1720000.00 | 220000.00  | Finalised | 2026-07-25 | Q3
404    | Unknown car              | Fathima Baker    | Mohammed Asif    | 1680000.00 | 230000.00  | Finalised | 2026-10-05 | Q4
405    | Mahindra Thar Roxx AX7L  | Unknown customer | Sreekanth Pillai | 2100000.00 | 250000.00  | Quote     | 2026-11-14 | Q4
407    | Unknown car              | Naveen Nair      | Suresh Kumar     | 1500000.00 | 50000.00   | Finalised | 2026-09-27 | Q3
408    | Hyundai Creta SX(O)      | Fathima Baker    | Suresh Kumar     | 1710000.00 | 210000.00  | Finalised | 2026-09-27 | Q3
409    | Toyota Innova Hycross ZX | Reetha Johny     | Suresh Kumar     | 2700000.00 | 300000.00  | Finalised | 2026-09-27 | Q3
410    | Maruti Suzuki Brezza     | Reetha Johny     | Suresh Kumar     | 1000000.00 | 260000.00  | Quote     | 2026-09-27 | Q3
412    | Kia Carens Clavis        | Naveen Nair      | Suresh Kumar     | 1138900.00 | 38900.00   | Finalised | 2026-09-30 | Q3
413    | Maruti Suzuki Swift ZXi+ | Fathima Rafeeq   | Suresh Kumar     | 800000.00  | 50000.00   | Finalised | 2026-09-30 | Q3

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

# Invalid Combinations for Transactions and its validations:

### Option1: Enter a duplicate transaction ID, throws error messages ‘Transaction ID already in use’

```
Select an Option: 1
Enter a Transaction ID: 401
Transaction ID already in use

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

## Option 2: Finalise a Quote – Invalid Quote ID – Alphabets

```
Select an Option: 2
                                        OPEN QUOTES
=============================================================================================
Quote ID | Car                     | Customer         | Employee         | Sale Price | Date
---------+-------------------------+------------------+------------------+------------+-----------
410      | Maruti Suzuki Brezza    | Reetha Johny     | Suresh Kumar     | 1000000.00 | 2026-09-27
405      | Mahindra Thar Roxx AX7L | Unknown customer | Sreekanth Pillai | 2100000.00 | 2026-11-14
Enter the Quote ID to finalise: ert
Invalid Input - Quote ID must be numerical.

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

## Option 2: Finalise a Quote – Invalid Quote ID numbers

```
Select an Option: 2
                                        OPEN QUOTES
=============================================================================================
Quote ID | Car                     | Customer         | Employee         | Sale Price | Date
---------+-------------------------+------------------+------------------+------------+-----------
410      | Maruti Suzuki Brezza    | Reetha Johny     | Suresh Kumar     | 1000000.00 | 2026-09-27
405      | Mahindra Thar Roxx AX7L | Unknown customer | Sreekanth Pillai | 2100000.00 | 2026-11-14
Enter the Quote ID to finalise: 999
Quote ID not found in the open quote list.

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

## Option 3: Cancel a Reservation – Invalid Quote ID - Alphabets

```
Select an Option: 3
                                        OPEN QUOTES
=============================================================================================
Quote ID | Car                     | Customer         | Employee         | Sale Price | Date
---------+-------------------------+------------------+------------------+------------+-----------
410      | Maruti Suzuki Brezza    | Reetha Johny     | Suresh Kumar     | 1000000.00 | 2026-09-27
405      | Mahindra Thar Roxx AX7L | Unknown customer | Sreekanth Pillai | 2100000.00 | 2026-11-14
Enter the Quote ID to cancel: rty
Invalid Input - Quote ID must be numerical.

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

## Option 3: Cancel a Reservation – Invalid Quote ID numbers

```
Select an Option: 3
                                        OPEN QUOTES
=============================================================================================
Quote ID | Car                     | Customer         | Employee         | Sale Price | Date
---------+-------------------------+------------------+------------------+------------+-----------
410      | Maruti Suzuki Brezza    | Reetha Johny     | Suresh Kumar     | 1000000.00 | 2026-09-27
405      | Mahindra Thar Roxx AX7L | Unknown customer | Sreekanth Pillai | 2100000.00 | 2026-11-14
Enter the Quote ID to cancel: 607
Quote ID not found in the open quote list.

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

### Option 4: View/Reprint an Invoice – Invalid Quote ID – Alphabets

```
Select an Option: 4
Enter the Transaction ID: tyu
ERROR: Transaction ID must be Numerical

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

### Option 4: View/Reprint an Invoice – Invalid Quote ID numbers

```
Select an Option: 4
Enter the Transaction ID: 509
Invalid Transaction ID

[1] Create a New Quote/Transaction
[2] Finalise a Quote
[3] Cancel a Reservation
[4] View/Reprint an Invoice
[5] Show All Transactions
[6] Back
Select an Option:
```

### Option 6: Back option will navigate back to main menu

```
Select an Option: 6

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option:
```

# Staff Details

### Option 5: Staff Details

```
       Select an option: 5

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

### Option 2: View all Staff

```
Select an Option: 2
                                    STAFF
================================================================================
Employee ID | Name              | Role    | Phone      | Salary   | Experience
------------+-------------------+---------+------------+----------+-----------
23          | Suresh Kumar      | Manager | 9089785678 | 78.00    | 90.00
101         | Mathew Thomas     | Manager | 9846011223 | 85000.00 | 7.50
102         | Mohammed Asif     | Sales   | 9745122334 | 38000.00 | 3.00
103         | Sreekanth Pillai  | Sales   | 9633233445 | 42000.00 | 4.50
104         | Anju Varghese     | Sales   | 9447344556 | 32000.00 | 1.50

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

## Option 3: Update Staff Details

```
Select an Option: 3
                                    STAFF
================================================================================
Employee ID | Name              | Role    | Phone      | Salary   | Experience
------------+-------------------+---------+------------+----------+-----------
23          | Suresh Kumar      | Manager | 9089785678 | 78.00    | 90.00
101         | Mathew Thomas     | Manager | 9846011223 | 85000.00 | 7.50
102         | Mohammed Asif     | Sales   | 9745122334 | 38000.00 | 3.00
103         | Sreekanth Pillai  | Sales   | 9633233445 | 42000.00 | 4.50
104         | Anju Varghese     | Sales   | 9447344556 | 32000.00 | 1.50
Enter Employee ID to edit: 102
Enter column to update (name, passkey, role, phone, salary, experience): name
Enter new value for name: Mohammed A Asif
Employee record updated successfully.

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

## Option 2: View all Staff – View the updated details of Employee ID 102

```
Select an Option: 2
                                    STAFF
================================================================================
Employee ID | Name              | Role    | Phone      | Salary   | Experience
------------+-------------------+---------+------------+----------+-----------
23          | Suresh Kumar      | Manager | 9089785678 | 78.00    | 90.00
101         | Mathew Thomas     | Manager | 9846011223 | 85000.00 | 7.50
102         | Mohammed A Asif   | Sales   | 9745122334 | 38000.00 | 3.00
103         | Sreekanth Pillai  | Sales   | 9633233445 | 42000.00 | 4.50
104         | Anju Varghese     | Sales   | 9447344556 | 32000.00 | 1.50

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

## Option 1: New Staff

```
Select an Option: 1
Enter the Employee ID: 105
Enter the Name: Leela Santhosh
Enter the Passkey: admi123
Enter the Role: Sales
Enter the Phone: 9667897878
Enter the Salary: 79900
Enter the Experience: 7.6
Employee added successfully.

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

## Option 2: View all Staff will display newly added Staff

```
Select an Option: 2
                                    STAFF
================================================================================
Employee ID | Name              | Role    | Phone      | Salary   | Experience
------------+-------------------+---------+------------+----------+-----------
23          | Suresh Kumar      | Manager | 9089785678 | 78.00    | 90.00
101         | Mathew Thomas     | Manager | 9846011223 | 85000.00 | 7.50
102         | Mohammed A Asif   | Sales   | 9745122334 | 38000.00 | 3.00
103         | Sreekanth Pillai  | Sales   | 9633233445 | 42000.00 | 4.50
104         | Anju Varghese     | Sales   | 9447344556 | 32000.00 | 1.50
105         | Leela Santhosh    | Sales   | 9667897878 | 79900.00 | 7.60

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

### Option 4: Delete a Staff Member

```
Select an Option: 4
Enter Employee ID to delete: 105
Employee deleted successfully.

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

### Option 2: View all Staff – Deleted staff removed from the table

```
Select an Option: 2
                                    STAFF
================================================================================
Employee ID | Name              | Role    | Phone      | Salary   | Experience
------------+-------------------+---------+------------+----------+-----------
23          | Suresh Kumar      | Manager | 9089785678 | 78.00    | 90.00
101         | Mathew Thomas     | Manager | 9846011223 | 85000.00 | 7.50
102         | Mohammed A Asif   | Sales   | 9745122334 | 38000.00 | 3.00
103         | Sreekanth Pillai  | Sales   | 9633233445 | 42000.00 | 4.50
104         | Anju Varghese     | Sales   | 9447344556 | 32000.00 | 1.50

[1] New Staff
[2] View all Staff
[3] Update Staff Details
[4] Delete a Staff Member
[5] Back
Select an Option:
```

### Option 5: Back option will navigate back to main menu

```
Select an Option: 5

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option:
```

# Statistics

### Option 6: Statistics

```
       Select an option: 6

[1] Showroom Overview
[2] Quarterly Report
[3] Staff Performance
[4] Customer History
[5] Back
Select an Option:
```

### Option 1: Showroom Overview

```
Select an Option: 1

========== SHOWROOM STATISTICS ==========
Total Cars in Stock: 83
Total Inventory Value (Cost): ₹95960000.00
Total Registered Customers: 5
Total Staff Members: 5
=========================================


[1] Showroom Overview
[2] Quarterly Report
[3] Staff Performance
```

## Option 2: Quarterly Report

```
Select an Option: 2
========== QUARTERLY REPORT ==========
---------------------------------------
Quarter: Q1
Transactions: 1
Total Sale Price: 2750000.00
Total Profit: 350000.00
Average Profit: 350000.000000
---------------------------------------
Quarter: Q2
Transactions: 1
Total Sale Price: 880000.00
Total Profit: 130000.00
Average Profit: 130000.000000
---------------------------------------
Quarter: Q3
Transactions: 6
Total Sale Price: 9568900.00
Total Profit: 868900.00
Average Profit: 144816.666667
---------------------------------------
Quarter: Q4
Transactions: 1
Total Sale Price: 1680000.00
Total Profit: 230000.00
Average Profit: 230000.000000
---------------------------------------
Total Sales: 14878900.00
Total Profit: 1578900.00
=======================================
```

## Option 3: Staff Performance

```
Select an Option: 3
========== STAFF SALES REPORT ==========
----------------------------------------
Employee ID: 23
Name: Suresh Kumar
Experience: 90.00
Sales: 5
Total Profit: 648900.00
----------------------------------------
Employee ID: 102
Name: Mohammed A Asif
Experience: 3.00
Sales: 3
Total Profit: 710000.00
----------------------------------------
Employee ID: 103
Name: Sreekanth Pillai
Experience: 4.50
Sales: 1
Total Profit: 220000.00
----------------------------------------
========================================

[1] Showroom Overview
[2] Quarterly Report
[3] Staff Performance
[4] Customer History
[5] Back
Select an Option:
```

**Option 4: Customer History – Enter Customer ID will display Purchase History**

```
Select an Option: 4
Enter the Customer ID: 305

========== PURCHASE HISTORY ==========
CUSTOMER:  Reetha Johny
--------------------------------------
Txn ID: 409
Date: 2026-09-27
Make: Toyota
Model: Innova Hycross ZX
Price: 2700000.00
Status: Finalised
--------------------------------------
Txn ID: 410
Date: 2026-09-27
Make: Maruti Suzuki
Model: Brezza
Price: 1000000.00
Status: Quote
--------------------------------------
======================================

[1] Showroom Overview
[2] Quarterly Report
[3] Staff Performance
[4] Customer History
[5] Back
Select an Option:
```

**Option 5: Back option will navigate back to main menu**

```
Select an Option: 5

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option:
```

# Administrator Menu

**Option 7: Administrator Menu**

```
       Select an option: 7

[1] WIPE ALL DATA
[2] Insert Dummy Values
[3] Repair Tables
[4] Back
Select an Option:
```

**Option 1: WIPE ALL DATA – prompt for confirmation message / warning message, after confirmation all data cleared from DB**

```
Select an Option: 1
WARNING: This operation will erase ALL DATA contained within the database.

To confirm, type 'Yes, do as I say!': Yes, do as I say!
Table Values Cleared

[1] WIPE ALL DATA
[2] Insert Dummy Values
[3] Repair Tables
[4] Back
Select an Option:
```

## Recheck the data is wiped from tables

```
Select an Option: 4

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option: 1

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option: 2
No Data in Table


[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
```

## Option 2: Insert Dummy Values

```
Select an Option: 2
Dummy values have been inserted successfully
Dummy Admin Name: Mathew Thomas
Dummy Admin Password: admin123

[1] WIPE ALL DATA
[2] Insert Dummy Values
[3] Repair Tables
[4] Back
Select an Option: 
```

## Check the Dummy values are Inserted to the tables

```
       Select an option: 1

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option: 2
                                        CUSTOMERS
=============================================================================================================
Customer ID | Name            | Phone      | Email                    | Address
------------+-----------------+------------+--------------------------+----------------------------
301         | Naveen Nair     | 9847123456 | naveen.nair@gmail.com    | Panampilly Nagar, Kochi
302         | Fathima Rafeeq  | 9447234567 | fathima.r@yahoo.com      | Manacaud, Thiruvananthapuram
303         | Rony Mathew     | 9744345678 | rony.mathew@outlook.com  | Kanjikuzhy, Kottayam
304         | Aisha Sulaiman  | 9895456789 | aisha.sulaiman@gmail.com | Palayam, Kozhikode
305         | Harikrishnan V. | 9496567890 | hari.krishnan@gmail.com  | Chembukkav, Thrissur

[1] Add New Customer
[2] View All Customers
[3] Update Customer Details
[4] Delete Customer Details
[5] Back
Select an Option: 
```

```
       Select an option: 2

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option: 2
                                                INVENTORY
=================================================================================================================
Car ID | Make          | Model              | Year | Cost Price | Selling Price | Available | Reserved | Sold
-------+---------------+--------------------+------+------------+---------------+-----------+----------+-----
201    | Toyota        | Innova Hycross ZX  | 2025 | 2400000.00 | 2750000.00    | 3         | 1        | 2
202    | Hyundai       | Creta SX(O)        | 2025 | 1500000.00 | 1720000.00    | 4         | 0        | 3
203    | Mahindra      | Thar Roxx AX7L     | 2026 | 1850000.00 | 2100000.00    | 0         | 1        | 4
204    | Maruti Suzuki | Swift ZXi+         | 2025 | 750000.00  | 880000.00     | 5         | 0        | 4
205    | Tata          | Nexon EV Empowered | 2026 | 1450000.00 | 1680000.00    | 2         | 0        | 1

[1] Add a New Car
[2] View Inventory
[3] Update Car Details
[4] Delete a Car
[5] Back
Select an Option:
```

## Option 3: Repair Tables

```
Select an Option: 3
Table Creation/Repair Complete

[1] WIPE ALL DATA
[2] Insert Dummy Values
[3] Repair Tables
[4] Back
Select an Option:
```

## Option 4: Back option will navigate back to main menu

```
Select an Option: 4

==========CAR SHOWROOM DATABASE==========
|        PLEASE SELECT AN OPTION        |
|        [1] Customer Details           |
|        [2] Inventory Management       |
|        [3] Transactions               |
|        [4] Exit                       |
|        [5] Staff Details              |
|        [6] Statistics                 |
|        [7] Administrator Menu         |
+---------------------------------------+
       Select an option: 2
```

# CONCLUSION

The Car Showroom Database project provides a practical way to manage the main operations of a car showroom. It uses Python for its menu-driven interface and MySQL to store customer, staff, inventory, and transaction records. The system supports adding, viewing, updating, and deleting records, while also allowing staff to create and manage sales quotes, finalize transactions, print invoices, and review showroom statistics.

By tracking available, reserved, and sold vehicles, the project helps keep inventory information consistent as transactions progress. Its reports also make it easier for managers to review sales, profits, staff performance, and customer purchase history. Overall, this project demonstrates how a database-backed application can organize showroom data and simplify routine business tasks.

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